

Project Initialization and Planning Phase

Date	07 July 2026
Team ID	XXXXXXXXXX
Project Title	Personalized Networking Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Personalized Networking Assistant** is an AI-powered web application designed to help users prepare for professional and social networking events by generating personalized conversation starters. The system analyzes event descriptions using **DistilBERT**, generates context-aware conversation prompts using **GPT-2**, verifies factual information through the **Wikipedia API**, and maintains session history for future reference. This solution enhances networking confidence, improves communication, and ensures reliable AI-generated suggestions.

Project Overview	
Objective	Develop an AI-powered networking assistant that analyzes event information, generates personalized conversation starters, verifies facts using Wikipedia, and stores session history to improve networking experiences.
Scope	The project focuses on intelligent networking assistance by combining Natural Language Processing (NLP), Generative AI, and fact verification. It supports user profile management, event analysis, AI-generated conversation prompts, session tracking, and interaction logging.
Problem Statement	
Description	Many students and professionals struggle to start meaningful conversations during networking events due to limited preparation, lack of confidence, and insufficient knowledge about event topics or attendees. Existing tools provide generic suggestions and rarely offer personalized or fact-verified guidance.

Impact	Addressing these challenges improves user confidence, encourages meaningful professional interactions, reduces preparation time, and provides trustworthy AI-assisted networking support.
Proposed Solution	
Approach	The system uses DistilBERT to analyze event descriptions and extract key themes. Based on the user's profile and event context, GPT-2 generates personalized conversation starters. Generated content is verified using the Wikipedia API , while networking sessions and logs are stored for future access and continuous improvement.
Key Features	• AI-powered personalized conversation starter generation. • Event theme extraction using DistilBERT. • Context-aware responses using GPT-2. • Wikipedia-based fact verification. • User profile and networking session management. • Conversation history and system logging. • Interactive web interface built with Streamlit and FastAPI.

Resource Requirements

Hardware		
Resource Type	Description	Specification / Allocation
Computing Resources	CPU/GPU for AI model inference	Intel Core i5/i7 or equivalent, optional NVIDIA GPU
Memory	RAM for model execution	8 GB (Minimum), 16 GB Recommended
Storage	Application, datasets, and logs	20 GB Available SSD Storage

Software		
Resource Type	Description	Specification / Allocation
Frameworks	Backend and Frontend Frameworks	FastAPI, Streamlit
Libraries	AI/ML and NLP Libraries	Transformers, PyTorch, Hugging Face, Wikipedia API, Pandas, NumPy
Development Environment	IDE and tools	Visual Studio Code, Python Virtual Environment (venv), Git
Data		
Resource Type	Description	Specification / Allocation
Data	User profiles, event descriptions, AI-generated conversation starters, verification results, and session logs	Stored in JSON/CSV files or MongoDB (depending on deployment)